

REVIEW

Open Access



Intersectoral collaboration for strengthening infectious disease prevention and control in Nigeria: a narrative review

Idowu Peter Adewumi¹ , Queensly Opeyemi Adeyemi² , Abisola Esther Babatope¹ ,
Damola Olanipekun Ajisafe^{3*} and Kayode Olayiwola Adepoju¹

*Correspondence:

Damola Olanipekun Ajisafe
ajisafeoyindamola@yahoo.com
¹Department of Health Information
Management, Faculty of Allied
Health Sciences, University of
Medical Sciences, Ondo, Ondo
State, Nigeria
²Department of Pediatrics, State
Specialist Hospital, Akure, Ondo
State, Nigeria
³Department of Health Information
Management, Obafemi Awolowo
University Teaching Hospitals
Complex, Ile-Ife, Osun State, Nigeria

Abstract

Background Infectious diseases remain a major public health burden in Nigeria, straining limited health resources. Effective response requires intersectoral collaboration, engaging sectors like health, education, water, and sanitation, to address underlying determinants and promote sustainable prevention strategies, especially in the face of complex, systemic health challenges. This review critically examines intersectoral collaboration for infectious disease prevention and control in Nigeria, emphasising the need for a coordinated, adaptive, and sustainable approach involving all sectors.

Methods This review adopted a narrative approach, retrieving relevant research articles from PubMed, Directory of Open Access Journal (DOAJ) and Google Scholar. Websites of the World Health Organisation and the Nigeria Centre for Disease Control and Prevention, Federal Ministries of Health, and Information (Nigeria), were consulted for relevant reports on infectious diseases prevention and control in Nigeria. The initial search yielded 146 records out of which 51 were considered eligible and included in the review. The review adopted system theory, which offered an analytical framework to facilitate the understanding of the interrelated roles and dynamics between sectors.

Findings Findings reveal that intersectoral collaboration in Nigeria is hindered by fragmented governance, poor coordination, inadequate funding, and policy misalignment. However, enabling factors include shared goals, donor support, and crisis-driven collaboration. Systems Theory highlights the interconnectedness of sectors and the importance of adaptive feedback loops, reinforcing the need for integrated strategies to strengthen collaborative responses to infectious disease challenges in Nigeria.

Conclusion It was concluded that strengthening intersectoral collaboration through policy reforms, sustained funding, and community engagement is essential for Nigeria's infectious disease control. Applying systems thinking fosters resilient, coordinated actions that improve long-term health outcomes across sectors.

Keywords Infectious diseases, Intersectoral collaboration, Public health, Systems theory, Nigeria



1 Introduction

Infectious diseases remain a persistent threat to Nigeria's public health landscape, demanding urgent, coordinated responses. As outbreaks increasingly strain health systems and disrupt sectors like housing, sanitation, education, and the economy, isolated efforts by individual sectors have hindered comprehensive public health outcomes. Rather, a robust, intersectoral strategy is vital to confronting these complex, interconnected challenges effectively [1]. Nigeria continues to suffer greatly from diseases, including cholera, TB, malaria, and HIV/AIDS, which have a substantial negative impact on both health and the economy [2]. In 2024, Nigeria recorded 10,502 Cholera cases with 359 mortalities, giving a 3.3% case fatality rate (CFR), attributable to poor water and sanitation facilities [3]. Furthermore, TB continues to pose serious danger with an estimated 590,000 new infections and 245,000 deaths annually, exacerbated by unequal healthcare access and overcrowding [4].

Malaria continues to pose a threat to public health, with an estimated 68 million infections and 194,000 deaths annually, with environmental factors, Nigeria's tropical climate, and inadequate health services all contributing to the high prevalence [5]. While there has been some progress in the fight against HIV/AIDS, stigma, limited access to care, and a lack of public awareness have contributed to the approximately 1,400 new infections and 50,000 AIDS-related deaths that occurred in 2023 [6]. These diseases' combined effects continue to strain the nation's healthcare system, underscoring the shortcomings of a health-sector-only strategy and the demand for creative, inter-sectoral solutions [7].

The socioeconomic impact of infectious diseases necessitates the need for a more comprehensive response approach, as affected people frequently experience prolonged sickness, decreased productivity, and a higher financial burden on families and public health institutions [8]. Poor communities bear a disproportionate amount of these costs, exacerbating health inequities and perpetuating poverty cycles [9]. Given the interconnectedness of the health and other sectors, a cross-sectoral strategy is required for effective prevention and control. Better sanitation, for example, reduces the incidence of cholera, and education can decrease the spread of HIV/AIDS and malaria by changing behaviour [10].

Globally, several models of intersectoral collaboration offer instructive insights for Nigeria. The Health in All Policies (HiAP) strategy in South Africa promotes policy coherence and accountability by including health issues in areas such as education and transport [11]. It encourages accountability by incorporating health impact assessments into policy decisions across ministries, mandating reporting on health outcomes in non-health sectors, and establishing cross-sectoral governance structures that hold various departments jointly accountable for achieving health-related goals [11]. However, issues including inconsistent stakeholder participation and a lack of policy coherence make it difficult to implement HiAP in Nigeria [12]. Nigeria has also demonstrated its agreement with HiAP mandates, as captured in the recent national health policy goals that acknowledge active community participation in health planning, implementation, monitoring, and evaluation [13]. Similarly, the WHO Healthy Cities Network in Finland is a prime example of cooperative urban planning that enhances health by creating safer spaces and easier access to services [14]. While Nigeria has attempted to replicate this

framework, sustained results have been hindered by the absence of institutionalised feedback systems and adaptive policy structures [15].

In Nigeria, the Integrated Disease Surveillance and Response (IDSR) system was instrumental during the 2014 Ebola outbreak, particularly for surveillance activities, including contact tracing, case detection, and reporting [16]. However, the Emergency Operations Centres (EOCs) and Incident Management System (IMS) were the primary platforms for cross-sector collaboration and data sharing. These coordination frameworks permitted uniform oversight, faster response, and enhanced surveillance sensitivity. Sustaining this synergy requires consistent funding and institutional integration [17]. Furthermore, Public Private Partnerships (PPPs) can leverage private-sector capacity for public health, as demonstrated during the COVID-19 pandemic. However, these approaches run the risk of becoming short-term if structural alignment and continual stakeholder participation are not maintained [18].

Additionally, the Whole-of-Government and Whole-of-Society frameworks emphasise how important intersectoral governance is when dealing with difficult public health issues. While Nigeria's political unrest makes it difficult to implement these strategies [19], when tailored to the Nigerian context, they could enhance policy integration and long-term health outcomes. This review critically examines intersectoral collaboration for infectious disease prevention and control in Nigeria using Systems Theory to provide unique insights that are consistent with established institutions as well as practical, context-based recommendations based on recent empirical evidence. It depicts feedback loops and methodically synthesises barriers and enablers in the health, education, water/sanitation, and local government sectors.

2 Methods

2.1 Data sources accessed

To access eligible articles and reports for this review, academic databases including PubMed and Directory of Open Access Journal (DOAJ) were searched. Google scholar was accessed for a wider reach of academic and grey literature while verified websites of organisations like the Federal Ministry of Health (Nigeria), Federal Ministry of Information & National Orientation, Nigeria Centre for Disease Control (NCDC), World Health Organisation (WHO), and the Malaria Consortium were accessed to retrieve programmatic and policy reports. While academic databases provide rigorous, peer-reviewed empirical studies, government and WHO reports provide current official data and policy frameworks. Google Scholar assists in capturing grey literature and reports that might otherwise go unindexed. The flexibility of narrative reviews allows for the use of a wide range of sources, even though they are prone to bias [20].

2.2 Search strategy

The review adopted Boolean operators: (intersectoral OR multisectoral OR cross-sector) AND (infectious disease OR public health response OR disease prevention) AND (systems theory OR systems thinking) AND Nigeria to refine the search. MeSH phrases like “intersectoral collaboration” and “infectious disease/prevention and control” were utilised in PubMed, and DOAJ and Google Scholar used similar keyword combinations. To retrieve policy reports and official data, a manual search was conducted on the websites of the Federal Ministry of Health, NCDC, WHO, Malaria Consortium, and the Federal

Ministry of Information and National Orientation, utilising internal search tools and appropriate document menus. These strategies guaranteed that both peer-reviewed and grey literature were adequately captured for policy relevance [21].

2.3 Eligibility criteria

The study included empirical research, reviews, and policy or programmatic papers focusing on intersectoral or multisectoral collaboration for infectious disease prevention or control in Nigeria, published in English between 2010 and 2025. Additionally, exemplar international studies were selected if they addressed intersectoral mechanisms or systems theory methods relevant to Nigeria. Articles and reports without an intersectoral component, addressed non-infectious diseases, or provided no transferable insights were excluded. Grey literature from government, NGOs, and international organisations was purposefully included to collect real-world policy and practice evidence [22].

2.4 Search result

A total of 146 records were found during the initial search, with 61 from PubMed, 46 from DOAJ, 21 from Google Scholar, and 18 from organisational websites. After removing 19 duplicates, 127 unique objects remained. A total of 51 papers were excluded because they were not relevant after being screened by title and abstract. Another 25 were excluded after full-text screening due to a lack of substantial empirical or policy content. Finally, 51 documents, including peer-reviewed academic papers, official reports, and grey literature, were chosen for detailed study (Fig. 1).

2.5 Data synthesis

The data from the included records were synthesized narratively under three major themes, including current state of intersectoral collaboration in Nigeria, barriers to effective collaboration, and facilitators and opportunities for improved collaboration. Recommendations were therefore suggested based on the findings from the literature. The discussion was underpinned by systems theory, which views public health as an interrelated system that encompasses domains such as housing, sanitation, health-care, and education; it emphasises complexity, feedback loops, and interdependencies, providing robust lens for examining how intersectoral interventions impact infectious disease outcomes [23]. Overall, the application of ST enhances a comprehensive understanding of collaborative gaps and synergies, guiding the development of integrated public health plans and policies that are specific to Nigeria's complex infectious disease landscape [23].

3 Discussion

3.1 Current state of intersectoral collaboration in Nigeria

Intersectoral collaboration has become more important in Nigerian public health policies, especially when it comes to addressing infectious diseases that are exacerbated by socioeconomic position, education, and sanitation [10]. Many programs aim to solve these issues by uniting different sectors in coordinated action. For instance, partnerships between the health and education sectors have been formed in response to diseases like HIV/AIDS, when health organisations and schools work together to increase youth awareness and encourage preventative actions [24]. In addition to HIV/AIDS, primary

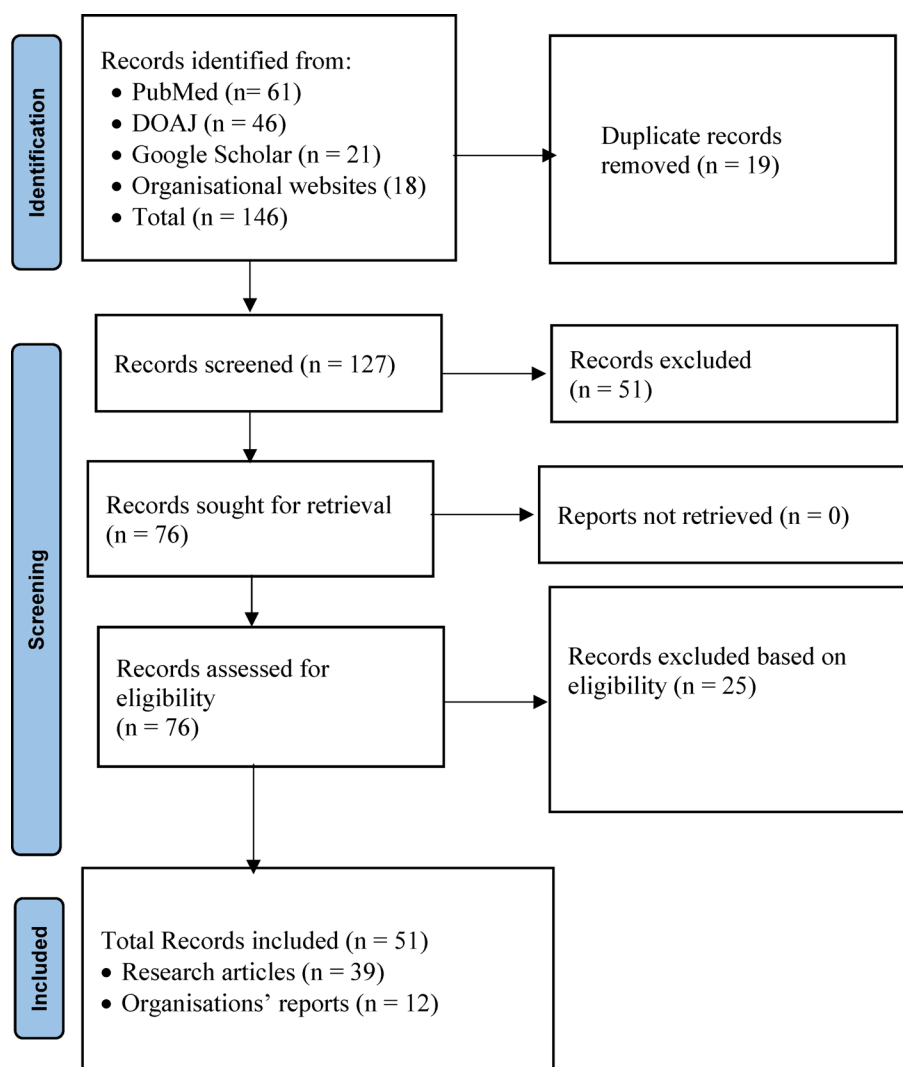


Fig. 1 PRISMA flowchart

school collaboration programs strive to integrate health education into the curriculum by addressing subjects like cleanliness to prevent cholera and other watery infections [25].

Recently, there has been collaboration between Nigeria's government, non-governmental organisations (NGOs), and community organisations to fight malaria through initiatives like the National Malaria Elimination Program (NMEP), which combines resources from the health, environmental, and local government sectors to improve vector control and community education [26]. Furthermore, outreach initiatives require community health workers who have been trained by both governmental and non-governmental organisations, especially in rural areas with poor access to medical treatment [27]. Through these intersectoral collaborations, government agencies' resources, community leaders' knowledge, and non-governmental organisations' reach are combined to reduce health inequities and provide access to preventive interventions [7].

Despite the concerted efforts, the outcomes of these collaborative undertakings have been diverse. Even though data from some programs indicate very slight long-term decreases in infection rates, they do indicate a rise in public awareness of illness and

preventative actions. For example, school-based programs have increased youth awareness of HIV/AIDS, but there have not been many behavioural changes, partly because of social and cultural barriers that still stand in the way of effective prevention [28]. Similarly, efforts to control malaria have shown potential in places with a lot of resources and community involvement, but these accomplishments usually end quickly when money is cut or governmental priorities change [29]. According to research, it can be difficult to determine the actual impact of intersectoral activities on public health outcomes in Nigeria since they usually lack uniform success criteria [30].

Intersectoral initiatives on TB interventions in Nigeria demonstrate a broader integration of disease control strategies. The National Tuberculosis and Leprosy Control Program (NTBLCP) promotes engagement with community partners, civil society, and NGOs to increase TB case detection, treatment adherence, and public knowledge of the disease [31]. NTBLCP specifically supports community outreach campaigns involving screening and awareness drives in local communities; operates a toll-free hotline (3340) that handles public enquiries and supports treatment adherence; and collaborates with youth-oriented groups such as the National Youth Service Corps (NYSC) for educational sessions and advocacy among young people [32].

To address stigma and accessibility challenges, NTBLCP works with global partners like the Global Fund and WHO to incorporate tuberculosis services into primary care [33]. While the goal of incorporating TB surveillance into the IDSR is to improve data exchange and epidemic monitoring, empirical research shows ongoing infrastructural and budget gaps in readiness. For example, according to a facility-readiness assessment conducted in the states of Oyo and Anambra, only about half of the facilities in Oyo and roughly 72.7% of those in Anambra had a steady power supply, and almost half (about 47.6%) had TB medicine stock-outs in the three months before the survey [34]. Additionally, according to a national report, just around 20% of Nigerian healthcare facilities can offer TB services, and only 24% of them can identify the disease [35]. These results imply that effective implementation of IDSR-TB integration is hampered by structural deficiencies in diagnostics, human resources, and stable funding, particularly in more rural or underdeveloped settings [36].

Ineffective interagency communication is a major obstacle to the effectiveness of these intersectoral collaborations, given that the environment, education, and health sectors usually function independently with little interaction, resulting in fragmented services and redundant work [37]. Financial constraints also provide a significant challenge because a large number of Nigerian health initiatives rely heavily on foreign funding, which is usually unstable and susceptible to changes in donor priorities [8]. While the efficacy of intersectoral activities may also be impacted by delays in their implementation caused by political unpredictability and administrative roadblocks, government policies may be unclear about roles and responsibilities across sectors, leading to inefficiencies, which is made worse by the absence of a coherent policy framework [19].

Based on a Systems Theory perspective that stresses interconnectedness and reciprocal accountability, Nigeria's experience highlights the need for a more structured approach to intersectoral collaboration [38]. In addition to improved coordination and communication, strengthening these connections calls for both a long-term financing source and more comprehensive evaluation frameworks to accurately measure effectiveness [19].

3.2 Barriers to effective collaboration

3.2.1 Structural barrier

Nigeria exhibits structural barriers to effective intersectoral collaboration, mostly attributed to disjointed health systems, inadequate finances, and poor inter-agency communication. However, the root cause of this health system fragmentation is the decentralised governance frameworks that distribute responsibilities across the federal, state, and local governments, frequently leading to disorganised efforts and resource duplication [39]. This fragmentation makes it difficult for health organisations to work together effectively with non-health sectors, particularly in areas like water and sanitation, making programs relating to disease prevention isolated, which reduces the efficacy of collaborative efforts [40].

Furthermore, cooperative efforts are hampered by inadequate communication across agencies. For example, it is challenging to coordinate initiatives that call for collaborative action because departments in charge of water and sanitation rarely maintain regular communication with health groups [41]. While deeper issues within the institutional frameworks governing public health in Nigeria are reflected in this lack of communication, uneven inter-agency communication exacerbates coordination challenges, often leading to wasted resource allocation, equal when efforts are made to close gaps [42]. Clear channels of communication between sectors are essential for intersectoral success, but they are still lacking in Nigeria [43].

Low funding is another significant structural barrier that limits the resources available to launch and maintain intersectoral projects. The health and non-health sectors in LMICs like Nigeria face financial constraints and usually depend on foreign donors instead of reliable domestic financing sources [44]. As evidenced in a case study on community health efforts in Nigeria, where budget cuts resulted in the cessation of essential services, this reliance on external finance makes intersectoral programs vulnerable to disintegration when donor support dwindles [45]. Systems theory, which highlights interconnection and feedback loops, suggests that sustainable funding should be a significant part of intersectoral initiatives because instability in one area can affect the resilience of the system as a whole [46].

3.2.2 Socio-cultural barriers

In addition to structural challenges, sociocultural barriers are a major factor in preventing cooperation and mistrust between communities and sectors is one such barrier. Nigerian sectors like health and education usually function independently due to historical trends towards independence and competition for few resources. Given the lack of trust created by this competitive strategy, authorities are compelled to prioritise sector-specific accomplishments over more general public health objectives [47]. This mistrust can be especially strong when sectors believe their resources or efforts may be diluted in a combined initiative, thereby hindering the kind of collaboration needed to address infectious disease challenges holistically [48].

Furthermore, sociocultural barriers are usually addressed by community engagement, which is necessary for effective collaboration to tailor public health initiatives to local needs [49]. However, Nigerian communities might not have the trust necessary to fully embrace intersectoral programs because of historical disenfranchisement and scepticism towards government interventions [50]. For instance, during the COVID-19 pandemic,

public hesitancy and a lack of adherence to health advice were caused by widespread concerns about vaccination safety and government transparency [51]. According to Systems Theory, which holds that input from all system components, including communities, is crucial for adaptive responses to public health needs, this scepticism emphasises the necessity of inclusive and transparent community engagement as a fundamental component of intersectoral collaboration [52].

Collaboration is further hampered by cultural customs and beliefs that are in direct opposition to public health initiatives. For instance, traditional beliefs frequently influence health practices in rural Nigeria, such as the reliance on herbal remedies rather than scientifically proven cures for illness prevention [53]. This cultural divide frequently restricts the implementation of preventive health initiatives since cooperation with sectors like education is essential for increasing health literacy and awareness but frequently lacks the cultural sensitivity necessary to be effective [54].

3.3 Facilitators and opportunities for improved collaboration

3.3.1 Policy-level facilitators

In Nigeria, one of the main initiatives that promotes intersectoral cooperation is the National Action Plan for Health Security (NAPHS), which, according to the World Health Organisation [55] is in line with the International Health Regulations (IHR) to improve health security. The focus of this strategy is on coordinating the efforts of the agriculture, health, and environmental sectors, especially in the areas of infectious disease preparedness and response [56]. Despite providing a comprehensive framework for cross-sector coordination, the NAPHS has challenges in implementing it, such as inadequate funding, uneven stakeholder participation, and frequent policy changes brought on by political turnover [57].

Strong local governance structures and the alignment of policies with popular aspirations are typically prerequisites for the sustainability of policy frameworks like the NAPHS. Building dynamic, feedback-rich lines of communication between local stakeholders and policymakers may increase the responsiveness of such initiatives, as proposed by the Systems Theory, allowing for greater flexibility in collaborating in response to new health hazards [58]. These cooperative efforts are directly supported by bolstering interagency accountability frameworks and guaranteeing budget distribution, which addresses shortcomings noted by scholars in their assessment of intersectoral coordination in Nigerian health policy [59]. Regular policy reviews conducted through multi-stakeholder feedback sessions may also help create a more self-correcting and sustainable framework, a prerequisite for long-term cooperative success [60].

Other programs that support intersectoral collaboration include Nigeria's Basic Health Care Provision Fund (BHCPF), which aims to provide healthcare to underprivileged communities. Although the health sector is the primary focus of the BHCPF [61], its implementation entails cooperation with finance, local government, and community leaders to guarantee fair resource distribution [62]. However, the effectiveness of BHCPF is hampered by delayed disbursement and partial funds transfers. Since its creation, only ₦56 billion out of the approved ₦89 billion has been released to states and FCT, benefiting around 7,250 primary healthcare centers out of over 35,500 [63]. It is therefore essential to strengthen joint monitoring and increase transparency in allocation and execution. In Systems Theory, feedback loops which are mechanisms that feed-back outputs

such as monitoring results or stakeholder input into decision-making, can help modify resource flows across sectors, particularly in response to critical public health requirements [64].

3.3.2 *Community-based facilitators*

To bridge the gaps between local demands and the national health agenda, non-governmental organisations (NGOs) and community-based organisations (CBOs) are crucial. In rural locations, institutions like the Community Health Practitioners Registration Board of Nigeria and several local NGOs have made it feasible to promote immunisation programs, disease preventive initiatives, and hygiene education [65]. These institutions, which typically support broader intersectoral engagement where the environmental, health, and education sectors work together to address regional public health challenges, typically act as channels for fostering community trust and involvement [66].

NGOs that engage the community, like the Malaria Consortium's integrated malaria control program, show how intersectoral collaboration can address public health issues in ways that are suitable for the local culture [67]. For instance, these programs have contributed to the reduction in the prevalence of malaria in many Nigerian states by fostering collaboration between the health sector and local educational authorities to raise awareness and teach preventative measures through partnerships with local health providers [68]. However, financial constraints, inadequate training for regional health professionals, and disjointed communication with higher-level officials usually hinder these exchanges [69]. According to systems theory, these NGOs are crucial components of a larger health system, and well-coordinated feedback loops between communities and legislators can close these gaps and increase program responsiveness to local needs [58].

In Nigeria, community-level facilitators face challenges in sustaining engagement and attaining scale, notwithstanding their contributions, and to ensure that programs stay effective and relevant, Systems Theory emphasises the need for flexible frameworks where ongoing feedback from community stakeholders may spur strategic policy changes [70]. A more unified approach to public health may be promoted by creating these feedback-rich platforms, which would enable NGOs and CBOs to interact more closely with government health initiatives.

4 Conclusion

This review underscores the vital role of intersectoral collaboration in addressing Nigeria's infectious disease burden, showing that collaborative solutions outside the health sector are necessary due to the intricate and interwoven nature of disease determinants, which include education, sanitation, water, housing, and health. Despite the potential of programs like the National Health Policy and collaborations between the health and education sectors, obstacles still exist, including sociocultural aversion to cooperation, inadequate funding, disjointed communication channels, and disordered institutional frameworks. In line with the system theory, this review promotes a systemic approach that acknowledges the importance of sectoral interdependencies, feedback loops, and adaptive processes in enhancing public health resilience. When adapted to local conditions, intersectoral policies are not only viable but also effective. Nigeria must therefore adopt comprehensive policy reforms, improve sectoral cooperation, and encourage continual community engagement to make significant progress.

5 Limitations of the study

During the review, some limitations were identified, including the study's dependence on narrative synthesis, which is prone to selection bias, and it does not follow the methodical rigour of a structured review and meta-analyses [71]. Furthermore, the findings may not be generalisable to other contexts due to the contextual focus on Nigeria. However, these limitations do not affect the outcome of the study, since the narrative approach enabled a comprehensive interpretation of a variety of evidence, and the research articles and reports accessed for this review offered trustworthy information. Despite the limited methodological scope, the integration of Systems Theory further reinforced a strong analytical lens, allowing for a greater comprehension of intersectoral dynamics crucial to the control of infectious diseases.

6 Recommendations

To improve intersectoral collaboration in infectious disease prevention, the following recommendations are given priority for implementation in Nigeria based on the findings and the application of Systems Theory. These initiatives ensure viability and sustainability by building upon established frameworks including the National Action Plan for Health Security (NAPHS), Emergency Operations Centres (EOCs), and the Basic Health Care Provision Fund (BHC PF) gateway.

6.1 Strengthening governance, monitoring and feedback through existing structures

Authorities like the National Primary Health Care Development Agency (NPHCDA) and State PHC Development Agencies should expand on the BHC PF gateway evaluation that was carried out in six Northern states (Bauchi, Borno, Kaduna, Kano, Sokoto, and Yobe). This evaluation found that early start-up activities, established coordination structures, and financial management systems were linked to success, but that implementation was hindered by inadequate supervision and late submission of business plans [72]. States should institute performance benchmarks (such as timely approval of business plans, ward PHC accreditation, and supervisory indicators) before disbursing cash. To ensure that non-health sectors (such as water, education, and local government) are held accountable for the responsibilities outlined in multi-sector workplans, the NAPHS Secretariat (directed by the NCDC/IHR-Focal Point) should incorporate intersectoral collaboration indicators into its annual reporting platform [73].

6.2 Increasing sustainable financing and transparent resource flow

The federal and state ministries of finance together with NPHCDA, should establish regulations that ensure a fixed portion of state health budgets is earmarked for intersectoral prevention programs (such as education outreach and water sanitation). Using evidence from BHC PF gateway states, funds should be disbursed only after fulfillment of clear preconditions (e.g. facility accreditation, valid business plans), to reduce delays and ensure resources reach PHC facilities promptly. To reward well-performing states, the Ministry of Budget & National Planning should also encourage matching funds or conditional allocations. Additionally, civil society participation in supervision through state supervision Committees should be mandatory to promote transparency. For instance, Rwanda has successfully defined roles across sectors (health, agriculture, and environment) and established clear surveillance and response protocols through its One Health

Strategic Plans and Multisectoral Coordination Mechanism. This has allowed for faster detection and more coordinated action [74, 75].

6.3 Expanding and institutionalising role of EOCs/emergency protocols for prevention

In addition to being designated as outbreak response centres, EOCs at the national and subnational levels (state PHEOCs) should be developed as continuous coordination centres for surveillance, data sharing, and cross-sector prevention. To detect early outbreaks of cholera or malaria, EOCs can pool surveillance data from education, water and sanitation, environmental health, and IDSR sectors. The Federal Ministry of Health, in partnership with the States, should release guidelines outlining the roles of each sector within EOC missions and guaranteeing financing for their continuous operation. To achieve this, Nigeria can learn from Uganda's PHEOC, created in 2013 and serves as a permanent multisectoral coordination centre in addition to responding to outbreaks. This is done by integrating surveillance data (public health, labs, and environmental health), ensuring stakeholder alignment and legal authority, and promoting continuous preparedness [76].

6.4 Capacity building and research

Stakeholders, including NPHCDA, and academic and research institutions should conduct comparative studies between well-performing states under the BHCPF gateway and those lagging. This is important to identify strategies that facilitate effective coordination, funding, and supervision. There should be training programs focusing on PHC facility managers, state and local government health officials, and non-health sector actors (such as those in the water, sanitation, and education sectors), on intersectoral planning, data use, and feedback loop mechanisms. Overall, outcomes of these evaluations should inform policy adjustment under NAPHS and BHCPF.

Acknowledgements

Not applicable.

Author contributions

IPA and QOA conceived the idea of the study; IPA, QOA, AEB, DOA, and KOA searched and retrieved research articles; IPA, QOA, AEB, DOA and KOA wrote the manuscript. All authors reviewed and approved the manuscript for publication.

Funding

No funding received.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 24 April 2025 / Accepted: 1 December 2025

Published online: 06 January 2026

References

- Such E, Smith K, Woods HB, Meier P. Governance of intersectoral collaborations for population health and to reduce health inequalities in High-Income countries: a complexity-informed systematic review. *Int J Health Policy Manag.* 2022;11(12):2780–92.
- Nwabuko C, Nnoli AM, Nwabuko CO, Nnaji OT. HIV/AIDS in Nigeria: risk assessment and the risk management. *ARC J Hematol.* 2019;4(2):8–14.
- Nigeria Centre for Disease Control and Prevention, Cholera Situation R. 2024 [cited 2024 Oct 10]. Available from: <http://ncdc.gov.ng/diseases/sitreps/?cat=7-name=An/20update/20of/20Cholera/20outbreak/20in/20Nigeria>
- Olaleye SA, Balogun OS, Adusei-Mensah F. Bibliometric structured review of tuberculosis in Nigeria. *Afr Health Sci.* 2023;23(2):139–60.
- World Health Organization. Report on malaria in Nigeria 2022 | WHO | Regional Office for Africa. 2022 [cited 2024 Oct 10]. Available from: <https://www.afro.who.int/countries/nigeria/publication/report-malaria-nigeria-2022-0>
- National Agency for the Control of Aids. NACA Nigeria. 2024 [cited 2024 Oct 11]. NACA reports 1,400 new HIV cases per week in Nigeria for 2023. Available from: <https://naca.gov.ng/naca-reports-1400-new-hiv-cases-per-week-in-nigeria-for-2023/>
- Rajabi M, Ebrahimi P, Aryankhesal A. Collaboration between the government and nongovernmental organizations in providing health-care services: A systematic review of challenges. *J Educ Health Promotion.* 2021;4(4):242.
- Abubakar I, Dalglish SL, Angell B, Sanuade O, Abimbola S, Adamu AL, et al. The Lancet Nigeria commission: investing in health and the future of the Nation. *Lancet Elsevier.* 2022;299:1155–200.
- Deng L, Le, Zhao F, Li ZW, Zhang WW, He GX, Ren X. Epidemiological characteristics of tuberculosis incidence and its macro-influence factors in Chinese Mainland during 2014–2021. *Infect Dis Poverty.* 2024;13(1):1–15.
- Ramponi F, Ssenyonjo A, Banda S, Aliti T, Nkhoma D, Kaonga O, et al. Demands for intersectoral actions to meet health challenges in East and Southern Africa and methods for their evaluation. *Value in Health Regional Issues.* 2024;39:74–83.
- Michel J, Chimbindi N, Mohlakoana N, Orgill M, Bärnighausen T, Obrist B, et al. How and why policy-practice gaps come about: a South African Universal Health Coverage context. *Journal of Global Health Reports.* 2019;3:e2019069.
- Aregbesola B. Health care in Nigeria: Challenges and recommendations. *Socialprotection.org.* 2019. pp. 1–7.
- Etiaba E, Agwu P, Conteh L, Onwujekwe O. Multisectoral contributions to health security and formal policy availability at the community level in Nigeria. *Front Public Health.* 2025;13:1505383.
- Tiirinki H, Sulander J, Sinervo T, Halme S, Keskimäki I. Integrating Health and Social Services in Finland: Regional Approaches and Governance Models. *Int J Integr Care.* 2022;22(3):18.
- Okeke C, Ezenwaka U, Ekenna A, Onyedinma C, Onwujekwe O. Analysing the progress in service delivery towards achieving universal health coverage in Nigeria: a scoping review. *BMC Health Serv Res.* 2023;23(1):1094.
- Ibrahim LM, Stephen M, Okudo I, Kitgaka SM, Mamadu IN, Njai IF, et al. A rapid assessment of the implementation of integrated disease surveillance and response system in Northeast Nigeria, 2017. *BMC Public Health.* 2020;20(1):1–8.
- Shuaib F, Gannala R, Musa EO, Mahoney FJ, Oguntimehin O, Nguku PM, et al. Ebola virus disease outbreak - Nigeria, July–September 2014. *MMWR Morb Mortal Wkly Rep.* 2014;63(39):867–72.
- Nederhand J, Klijn EH. Stakeholder involvement in Public–Private partnerships: its influence on the innovative character of projects and on project performance. *Adm Soc.* 2019;51(8):1200–26.
- Watkins DA, Nugent R, Saxenian H, Yamey G, Danforth K, González-Pier E, et al. Intersectoral policy priorities for health. In: Jamison DT, Gelband H, Horton S, et al., editors. *Disease control Priorities, improving health and reducing poverty. Volume 9. The International Bank for Reconstruction and Development / The World Bank;* 2017. pp. 23–41.
- Lisy K, Porritt K. Narrative synthesis. *Int J Evid Based Healthc.* 2016;14(4):201.
- McGowan J, Sampson M, Salzweid DM, Cogo E, Foerster V, Lefebvre C. PRESS peer review of electronic search strategies: 2015 guideline statement. *J Clin Epidemiol.* 2016;75:40–6.
- Patino CM, Ferreira JC. Inclusion and exclusion criteria in research studies: definitions and why they matter. *J Bras Pneumol.* 2018;44(2):84.
- Ellwanger JH, da Veiga ABG, Kaminski VdeL, Valverde-Villegas JM, de Freitas AWQ, Chies JAB. Control and prevention of infectious diseases from a one health perspective. *Genet Mol Biol.* 2021;44(1):1–23.
- Arije O, Udoh E, Ijadunola K, Afolabi O, Aransiola J, Omoregie G, et al. Combination prevention package of interventions for reducing vulnerability to HIV among adolescent girls and young women in Nigeria: an action research. *PLoS One.* 2023. <https://doi.org/10.1371/journal.pone.0279077>.
- Daka H, Jacob J, Simeon M, Mulenga-Hagane LM, Kalisto K, Masaiti G et al. Integration of HIV and AIDS into primary curriculum: teacher training curriculum. *Int J Humanit Soc Sci Educ.* 2021;8(2).
- Federal Ministry of Health Nigeria. National Malaria Elimination Programme (NMEP). *Health.Gov.Ng.* 2022 [cited 2024 Nov 8]. Available from: <https://site.health.gov.ng/nmep>
- Javanparast S, Windle A, Freeman T, Baum F. Community health worker programs to improve healthcare access and equity: are they only relevant to low- and middle-income countries? *Int J Health Policy Manag.* 2018;7(10):943–54.
- Ezelote CJ, Osuji NJ, Mbachu AJ, Odinaka CK, Okwuosa OM, Oli CJ, et al. Effect of peer health education intervention on HIV/AIDS knowledge amongst in-school adolescents in secondary schools in Imo State, Nigeria. *BMC Public Health.* 2024. <https://doi.org/10.1186/s12889-024-18536-4>.
- Oyibo W, Ntadom G, Uhomiohi P, Oresanya O, Ogbulafor N, Ajumobi O, et al. Geographical and temporal variation in reduction of malaria infection among children under 5 years of age throughout Nigeria. *BMJ Glob Health.* 2021;6(2):e004250.
- van Dale D, Lemmens L, Hendriksen M, Savolainen N, Nagy P, Marosi E, et al. Recommendations for effective intersectoral collaboration in health promotion interventions: results from joint action chrodis-plus work package 5 activities. *Int J Environ Res Public Health.* 2020;17(18):1–20.
- Federal Ministry of Health. National Strategic Plan for Tuberculosis Control. 2021 [cited 2024 Nov 15]. Available from: <https://ntblcp.org.ng/content/uploads/2023/06/NSP-For-TB-CONTROL-2021-2025-compressed.pdf>
- Federal Ministry of Health. 2023 Annual Tuberculosis Report. 2023 [cited 2024 Nov 15]. Available from: https://ntblcp.org.ng/content/uploads/2025/02/NTBLCP-2023-Annual-Report-wecompress.com_.pdf
- WHO Africa, finding in Nigeria. Intensifying new initiatives for TB case- 2024 [cited 2024 Nov 15]. Available from: <https://www.afro.who.int/countries/nigeria/news/intensifying-new-initiatives-tb-case-finding-nigeria>

34. Oluwasanu MM, Hassan A, Adebayo AM, Ogbuji QC, Adeniyi BO, Adewole DA, et al. General and tuberculosis-specific service readiness in two States in Nigeria. *BMC Health Serv Res*. 2020;20(1):1–15.
35. Ogunniyi TJ, Abdulganiyu MO, Issa JB, Abdulhameed I, Batisani K, BMJ Publishing Group. Ending tuberculosis in Nigeria: a priority by 2030. *BMJ Glob Health*. 2024;9:e016820.
36. Ogbuabor DC, Onwujekwe OE. Governance of tuberculosis control programme in Nigeria. 2019;2:1–11.
37. Auschra C. Barriers to the integration of care in inter-organisational settings: a literature review. *Int J Integr Care Ubiquity Press Ltd*. 2018;18:1–14.
38. Valero-Silva N, Systems, Theory. Implementation of SDGs. Cham: In Springer; 2020. pp. 840–8.
39. Ajisegiri WS, Abimbola S, Tesema AG, Odusanya OO, Ojji DB, Peiris D, et al. Aligning policymaking in decentralized health systems: evaluation of strategies to prevent and control non-communicable diseases in Nigeria. *PLoS Glob Public Health*. 2021;1(11):e0000050.
40. Croke K, Ogbuaji O. Health reform in Nigeria: the politics of primary health care and universal health coverage. *Health Policy Plan*. 2024;39(1):22–31.
41. Freeman MC, Ogden S, Jacobson J, Abbott D, Addiss DG, Amnie AG, et al. Integration of water, sanitation, and hygiene for the prevention and control of neglected tropical diseases: a rationale for inter-sectoral collaboration. *PLoS Negl Trop Dis*. 2013. <https://doi.org/10.1371/journal.pntd.0002439>.
42. Frieden TR. Six components necessary for effective public health program implementation. *Am J Public Health*. 2014;104(1):17–22.
43. Igwe MC, Obeagu EI. The scopes and implications of health communication in public health. *Practices Nigeria*. 2024;11(1):1–7.
44. Adebisi YA, Umah JO, Olaoye OC, Alaran AJ, Sina-Odunsi AB III. DELP. Assessment of health budgetary allocation and expenditure toward achieving universal health coverage in Nigeria. *Int J Heal Life Sci*. 2020;6(2):e102552.
45. Obi US, Ogbuaji O, Mao W, Shahid M, Onwujekwe O, Yamey G. Progress in the face of cuts: a qualitative Nigerian case study of maintaining progress towards universal health coverage after losing donor assistance. *Health Policy Plan*. 2021;36(7):1045–57.
46. Morgan MJ, Stratford E, Harpur S, Rowbotham S. A systems thinking approach for community health and wellbeing. *Syst Pract Action Res*. 2024;37(2):161–83.
47. Onolememen MO. The impact of leadership on the governance of infrastructure development in Nigeria. *Walden University*; 2015.
48. Uchejeso OM, Etukudoh NS, Chongs ME, Ime DM. Challenges of inter-professional teamwork in Nigeria healthcare. *Www. Intechopen.Com*. 2022;12.
49. Saito K, Komasaawa M, Ssekitooleko R, Aung MN. Enhancing community health system resilience: lessons learnt during the COVID-19 pandemic in Uganda through the qualitative inquiry of the COVID Task Force. *Front Public Health*. 2023. <https://doi.org/10.3389/fpubh.2023.1214307>.
50. Hussein H. Prevalence of diarrhea and associated risk factors in children under five years of age in Northern Nigeria: a secondary data analysis of Nigeria demographic and health survey 2013. *Nord J Afr Stud*. 2017;16(1):64–74.
51. Ojo TO, Ojo AO, Ojo OE, Akinwalere BO, Akinwumi AF. Determinants of COVID-19 vaccine uptake among Nigerians: evidence from a cross-sectional national survey. *Arch Public Health*. 2023. <https://doi.org/10.1186/s13690-023-01107-1>.
52. Cordon CP. System theories: an overview of various system theories and its. *Application Healthc*. 2013;2(1):13–22.
53. Awasthi KR, Jancey J, Clements ACA, Sah RK, Koirala MP, Chalise B, et al. Traditional beliefs, practices, and migration: a risk to malaria transmission in rural Nepal. *Int J Environ Res Public Health*. 2022;19(24):16872.
54. Williams MS. A qualitative assessment of the social cultural factors that influence cervical cancer screening behaviors and the health communication preferences of women in Kumasi, Ghana. *J Cancer Educ*. 2014;29(3):555–62.
55. World Health Organisation. *2who.int*. 2024 [cited 2024 Nov 9]. p. 1 National Action Plan for Health Security. Available from: <https://www.who.int/emergencies/operations/international-health-regulations-monitoring-evaluation-framework/national-action-plan-for-health-security>
56. Nigeria Centre for Disease Control and Prevention. National Action Plan on Health Security. Nigeria Centre for Disease Control and Prevention. 2022 [cited 2024 Nov 9]. Available from: <https://ncdc.gov.ng/>
57. Ojo OE, Dalhat M, Garfield R, Lee C, Oyeibanji O, Oyetunji A, et al. Nigeria's Joint External Evaluation and National Action Plan for Health Security. *Health Secur*. 2020;18(1):16–20.
58. Mele C, Pels J, Polese F. A brief review of systems theories and their managerial applications. *Serv Sci*. 2010;2(1–2):126–35.
59. Amri M, Chatur A, O'Campo P. Intersectoral and multisectoral approaches to health policy: an umbrella review protocol. *Health Res Policy Syst BioMed Cent Ltd*. 2022;20:1–5.
60. Fobbe L, Hilletofth P. The role of stakeholder interaction in sustainable business models. A systematic literature review. *J Clean Prod*. 2021;327:129510.
61. Fabong HAM, Igboechesi GP. A review of basic health care provision fund (BHCF) and primary health care assessment in plateau state. *Int J Heal Psychol Res*. 2023;11(2):41–55.
62. Nwobodo E, Ukwuije F, Ekwuatu U, Ezugwu F, Nwobodo N, Ojiakor A. Assessment of the progress of the implementation of the basic health care provision fund in South East States of Nigeria. *Trop J Med Res*. 2022;21(1):75–85.
63. Federal Ministry of Information and National Orientation. BHCPF: FG Disburses N56 Billion to 36 States, FCT. 2022 [cited 2025 Sep 13]. Available from: <https://fmino.gov.ng/bhcpf-fg-disburses-n56-billion-to-36-states-fct/>
64. Farrell A, Hu M, Chin L, Evbuoma E, Liem W, Ballard E. Understanding systems from a feedback perspective. *Methods Br Ser*. 1.05: Systems Thinking Foundations. St. Louis (MO): Social System Design Lab; 2021. Available from: <https://doi.org/10.7936/t4k1-8g36>
65. Dotimi DA, Wamanyi Y, Silas JU. Community health practice: the Nigerian model. *Int J Community Med Public Health*. 2023;11(1):560–5.
66. Duru JI, Usman S, Adeosun O, Stamidis KV, Bologna L. Contributions of volunteer community mobilizers to polio eradication in Nigeria: the experiences of Non-governmental and civil society organizations. *Am J Trop Med Hyg*. 2019;101(4):74–84.
67. Awasthi KR, Jancey J, Clements ACA, Rai R, Leavy JE. Community engagement approaches for malaria prevention, control and elimination: a scoping review. *BMJ Open*. 2024;14(2):1–17.

68. Malaria Consortium. Support to the National Malaria Programme. 2020; Available from: <https://www.malariaconsortium.org/projects/sunmap2>
69. Ogbulafor N, Uhomoibhi P, Shekarau E, Nikau J, Okoronkwo C, Fanou NML, et al. Facilitators and barriers to seasonal malaria chemoprevention (SMC) uptake in Nigeria: a qualitative approach. *Malar J.* 2023;22(1):1–13.
70. Onwujekwe O, Mangham-Jefferies L, Cundill B, Alexander N, Langham J, Ibe O, et al. Effectiveness of provider and community interventions to improve treatment of uncomplicated malaria in Nigeria: a cluster randomized controlled trial. *PLoS One.* 2015;10(8):e0133832.
71. Tawfik GM, Dila KAS, Mohamed MYF, Tam DNH, Kien ND, Ahmed AM, et al. A step by step guide for conducting a systematic review and meta-analysis with simulation data. *Trop Med Health.* 2019;47(1):1–9.
72. Igbokwe U, Ibrahim R, Aina M, Umar M, Salihu M, Omoregie E, et al. Evaluating the implementation of the National primary health care development agency (NPHCDA) gateway for the basic healthcare provision fund (BHCPF) across six Northern States in Nigeria. *BMC Health Serv Res.* 2024;24(1):1404.
73. Nigeria Center for Disease Control. National Action Plan for Health Security 2018–2022 [Internet]. 2018. Available from: <https://ncdc.gov.ng/themes/common/files/establishment/5e88f9e22d2b4e4563b527005c8a0c43.pdf?>
74. United Nations. United Nations Rwanda. 2019 [cited 2025 Sep 14]. Rwanda gets new One Health strategic plan. Available from: <https://rwanda.un.org/en/3061-rwanda-gets-new-one-health-strategic-plan>
75. Adewumi IP, Babatope AE, Adepoju KO, Olaleye AM, Ajisafe DO, Okunlola OA. Trends, public health implications, and emerging concerns of Lassa fever in Ondo state, Nigeria. *Discov Public Health.* 2025;22(1):399.
76. Kayiwa J, Homsy J, Nelson LJ, Ocom F, Kasule JN, Wetaka MM, et al. Establishing a public health emergency operations center in an outbreak-prone country: lessons learned in Uganda, January 2014 to December 2021. *Health Secur.* 2022;20(5):394–407.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.